Theorem .1 (Approximate Restricted-Game Equivalence)**.** *Let*

$$B_1 = \{\pi_{1,1}, \dots, \pi_{1,n}\}, \qquad B_2 = \{\pi_{2,1}, \dots, \pi_{2,m}\}$$

be the current basis populations, and let the true restricted-game payoff matrix $M \in \mathbb{R}^{n\times m}$ be defined by

$$M_{ij} := J_S(\pi_{1,i}, \pi_{2,j}).$$

Let $\widetilde{M} \in \mathbb{R}^{n\times m}$ denote the synthesized payoff matrix produced by the learned abstraction, i.e.,

$$\widetilde{M}_{ij} := \widetilde{J}(\pi_{1,i}, \pi_{2,j}).$$

Assume that

$$\|M - \widetilde{M}\|_{\max} := \max_{i\in[n],\, j\in[m]} |M_{ij} - \widetilde{M}_{ij}| \le \varepsilon_M.$$

Suppose that $(\widehat{\alpha}, \widehat{\beta}) \in \Delta_n \times \Delta_m$ is an η-approximate Nash equilibrium of the matrix game induced by $\widetilde{M}$ in the sense that

$$\max_{\alpha\in\Delta_n} \alpha^\top \widetilde{M}\, \widehat{\beta} - \widehat{\alpha}^\top \widetilde{M}\, \widehat{\beta} \le \eta, \tag{1}$$

$$\widehat{\alpha}^\top \widetilde{M}\, \widehat{\beta} - \min_{\beta\in\Delta_m} \widehat{\alpha}^\top \widetilde{M}\, \beta \le \eta. \tag{2}$$

Define the restricted-game NashConv on a zero-sum payoff matrix $A \in \mathbb{R}^{n\times m}$ by

$$\mathrm{NashConv}_A(\alpha, \beta) := \Big(\max_{\alpha'\in\Delta_n} {\alpha'}^\top A\beta - \alpha^\top A\beta\Big) + \Big(\alpha^\top A\beta - \min_{\beta'\in\Delta_m} \alpha^\top A\beta'\Big).$$

Then

$$\mathrm{NashConv}_M(\widehat{\alpha}, \widehat{\beta}) \le 4\varepsilon_M + 2\eta.$$

In particular, if $(\widehat{\alpha}, \widehat{\beta})$ is an exact Nash equilibrium of the synthesized game induced by $\widetilde{M}$ (i.e., $\eta = 0$), then

$$\mathrm{NashConv}_M(\widehat{\alpha}, \widehat{\beta}) \le 4\varepsilon_M.$$

Proof. We first record a simple perturbation bound. For any $\alpha \in \Delta_n$ and $\beta \in \Delta_m$,

$$\begin{aligned}
\big|\alpha^\top (M - \widetilde{M})\beta\big| &= \left|\sum_{i=1}^{n}\sum_{j=1}^{m} \alpha_i\beta_j (M_{ij} - \widetilde{M}_{ij})\right| \\
&\le \sum_{i=1}^{n}\sum_{j=1}^{m} \alpha_i\beta_j |M_{ij} - \widetilde{M}_{ij}| \\
&\le \|M - \widetilde{M}\|_{\max} \sum_{i=1}^{n}\sum_{j=1}^{m} \alpha_i\beta_j \\
&= \|M - \widetilde{M}\|_{\max} \le \varepsilon_M.
\end{aligned} \tag{3}$$

We now bound the row-player deviation term in the true restricted game. For any $\alpha \in \Delta_n$, by (3),

$$\begin{aligned}
\alpha^\top M\widehat{\beta} &\le \alpha^\top \widetilde{M}\widehat{\beta} + \varepsilon_M \\
&\le \widehat{\alpha}^\top \widetilde{M}\widehat{\beta} + \eta + \varepsilon_M \qquad \text{(by (1))} \\
&\le \widehat{\alpha}^\top M\widehat{\beta} + \eta + 2\varepsilon_M.
\end{aligned} \tag{4}$$

Taking the maximum over $\alpha \in \Delta_n$ yields

$$\max_{\alpha\in\Delta_n} \alpha^\top M\widehat{\beta} - \widehat{\alpha}^\top M\widehat{\beta} \le \eta + 2\varepsilon_M.$$

Next we bound the column-player deviation term. For any $\beta \in \Delta_m$, again by (3),

$$\begin{aligned} \widehat{\alpha}^\top M\beta &\geq \widehat{\alpha}^\top \widetilde{M}\beta - \varepsilon_M \\ &\geq \widehat{\alpha}^\top \widetilde{M}\widehat{\beta} - \eta - \varepsilon_M \qquad \text{(by (2))} \\ &\geq \widehat{\alpha}^\top M\widehat{\beta} - \eta - 2\varepsilon_M. \end{aligned} \tag{5}$$

Taking the minimum over $\beta \in \Delta_m$ gives

$$\widehat{\alpha}^\top M\widehat{\beta} - \min_{\beta\in\Delta_m} \widehat{\alpha}^\top M\beta \leq \eta + 2\varepsilon_M.$$

Adding the two one-sided bounds proves that

$$\mathrm{NashConv}_M(\widehat{\alpha}, \widehat{\beta}) \leq (\eta + 2\varepsilon_M) + (\eta + 2\varepsilon_M) = 4\varepsilon_M + 2\eta.$$

The exact-equilibrium case follows by setting $\eta = 0$. □

Remark .2. The normalization above defines an "η-approximate Nash equilibrium" through two one-sided regret inequalities, each bounded by η. If instead one defines approximation directly by

$$\mathrm{NashConv}_{\widetilde{M}}(\widehat{\alpha}, \widehat{\beta}) \leq \eta,$$

then the same argument yields the slightly sharper bound

$$\mathrm{NashConv}_M(\widehat{\alpha}, \widehat{\beta}) \leq 4\varepsilon_M + \eta.$$

This result provides an operational bridge between the exact row-space theory and the practical learned setting. In particular, it justifies using restricted-game diagnostics such as $|M - \widetilde{M}|_{\max}$, MAE, and $\mathrm{NashConv}_M(\hat{\alpha}, \hat{\beta})$ to assess whether the learned abstraction is sufficiently accurate for meta-solving, even when a direct global test of row-space coverage is unavailable.

Importantly, this should be interpreted as a restricted-game certificate rather than a global certificate on the full transition subspace.

Besides the abstraction-dimension and sample-complexity analysis, we also diagnose how the learned abstraction behaves with respect to the row-space coverage requirement in practice. In the learned setting, directly checking the global condition $\mathrm{Row}(P_\nu) \supseteq \mathcal{F}$ is infeasible, especially in large games. We therefore evaluate an operational proxy at the restricted-game level: whether the synthesized payoff matrix $\widetilde{M}$ remains close to the reference restricted-game matrix M, and whether the meta-strategy computed from $\widetilde{M}$ remains low-error when evaluated on M.

Figure 1 reports these diagnostics on three tasks: Kuhn poker, Leduc poker, and Liar's dice. The first row shows the restricted-game matrix error over outer iterations, including both $|M - \widetilde{M}|_{\max}$ and MAE. The second row plots $|M - \widetilde{M}|_{\max}$ against the true restricted-game $\mathrm{NashConv}_M(\hat{\alpha}, \hat{\beta})$, where $(\hat{\alpha}, \hat{\beta})$ is the meta-strategy computed from the synthesized matrix $\widetilde{M}$. Through the approximate restricted-game equivalence theorem, these quantities directly measure whether abstraction error translates into degradation of the true restricted-game solution quality.

The purpose of this figure is not to certify the exact constant in the theorem, nor to claim global subspace recovery. Rather, it provides an empirical view of how the learned abstraction evolves during PSRO and whether it remains sufficiently accurate on the strategically relevant basis populations. This perspective is particularly important in larger tasks, where a global algebraic certificate is unrealistic and the practically meaningful question is whether the learned abstraction preserves the restricted-game payoffs and equilibrium quality encountered during training.

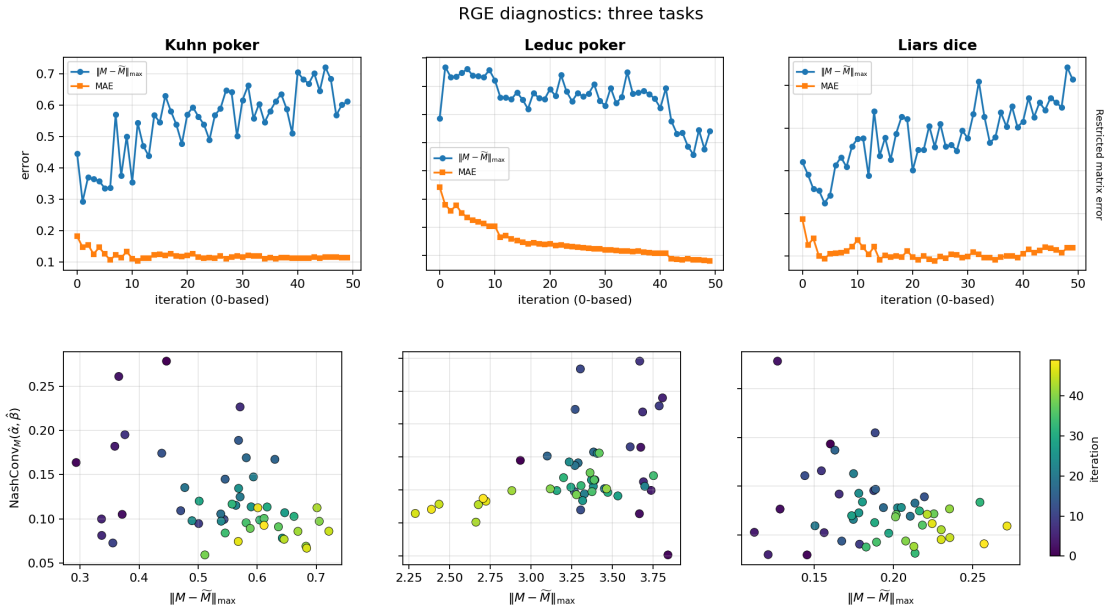

Figure 1. Diagnostics of approximate restricted-game equivalence on three tasks. Columns correspond to Kuhn poker, Leduc poker, and Liar's dice. In the first row, we plot the restricted-game payoff-matrix discrepancy between the true matrix M and the synthesized matrix $\widetilde{M}$ over outer iterations, including $|M - \widetilde{M}|_{\max}$ and MAE. In the second row, we plot the true restricted-game $\mathrm{NashConv}_M(\hat{\alpha}, \hat{\beta})$ against $|M - \widetilde{M}|_{\max}$, where $(\hat{\alpha}, \hat{\beta})$ is computed from $\widetilde{M}$. The figure provides an empirical diagnostic of whether the learned abstraction preserves restricted-game payoffs and equilibrium quality during training. It is intended as an operational proxy for approximate row-space coverage in the learned setting, rather than a direct certificate of the global condition $\mathrm{Row}(P_\nu) \supseteq \mathcal{F}$.